

Networks Assignment-1

GROUP 78

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TRACE FILE LINKS:

[DRIVE LINK](#)

QUESTION 1:

ALL THE PROTOCOLS USED BY **MICROSOFT TEAMS** ARE VISILBE IN THE BELOW SCREENSHOT OF STATISTICS->PROTOCOL HIERARCHY.

FROM HOSTEL ROOM ON MOBILE HOTSPOT:

Protocol	Percent Packets	Packets	Percent Bytes	Bytes	Bits/s	End Packets	End Bytes	End Bits/s	PDU's
▼ Frame	100.0	2373	100.0	1772504	120 k	0	0	0	2373
▼ Ethernet	100.0	2373	1.9	33223	2261	0	0	0	2373
▼ Internet Protocol Version 6	3.6	86	0.2	3440	234	0	0	0	86
▼ User Datagram Protocol	3.4	80	0.0	640	43	0	0	0	80
Domain Name System	3.4	80	1.0	16953	1154	80	16953	1154	80
▼ Transmission Control Protocol	0.3	6	0.0	120	8	0	0	0	6
Domain Name System	0.3	6	0.1	2035	138	6	2035	138	6
▼ Internet Protocol Version 4	96.4	2287	2.6	45740	3113	0	0	0	2287
▼ User Datagram Protocol	0.6	14	0.0	112	7	0	0	0	14
Domain Name System	0.6	14	0.2	3774	256	14	3774	256	14
▼ Transmission Control Protocol	95.8	2273	2.6	46096	3137	1705	34736	2364	2273
Transport Layer Security	23.9	566	95.9	1700075	115 k	566	700632	47 k	636
Data	0.1	2	0.0	2	0	2	2	0	2

FROM LAB ON ETHERNET:

Protocol	Percent Packets	Packets	Percent Bytes	Bytes	Bits/s	End Packets	End Bytes	End Bits/s	PDU's
▼ Frame	100.0	854	100.0	412396	48 k	0	0	0	854
▼ Ethernet	100.0	854	2.9	11956	1406	0	0	0	854
▼ Internet Protocol Version 4	100.0	854	4.1	17080	2009	0	0	0	854
▼ User Datagram Protocol	5.4	46	0.1	368	43	0	0	0	46
Domain Name System	5.4	46	2.2	9090	1069	46	9090	1069	46
▼ Transmission Control Protocol	94.6	808	4.0	16532	1945	502	10412	1225	808
Transport Layer Security	35.4	302	89.4	368618	43 k	302	345407	40 k	313
Domain Name System	0.2	2	0.2	676	79	2	676	79	2
Data	0.2	2	0.0	2	0	2	2	0	2

PROTOCOLS USED LAYER WISE:

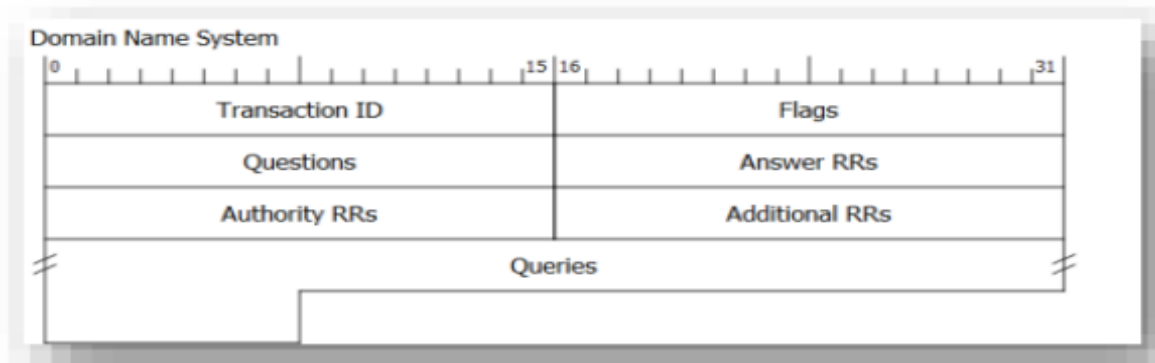
1) APPLICATION LAYER:

→DNS (DOMAIN NAME SYSTEM):

DNS, or the Domain Name System, translates human-readable domain names. DNS used the User Datagram Protocol (UDP) on port 53 to serve DNS queries. UDP is preferred because it is fast and has low overhead. A DNS query is a single UDP request from the DNS client followed by a single UDP reply from the server. DNS has two types of messages: query and response. Both types have the same format. The identification field is made up of 16 bits which are used to match responses with requests from the client side.

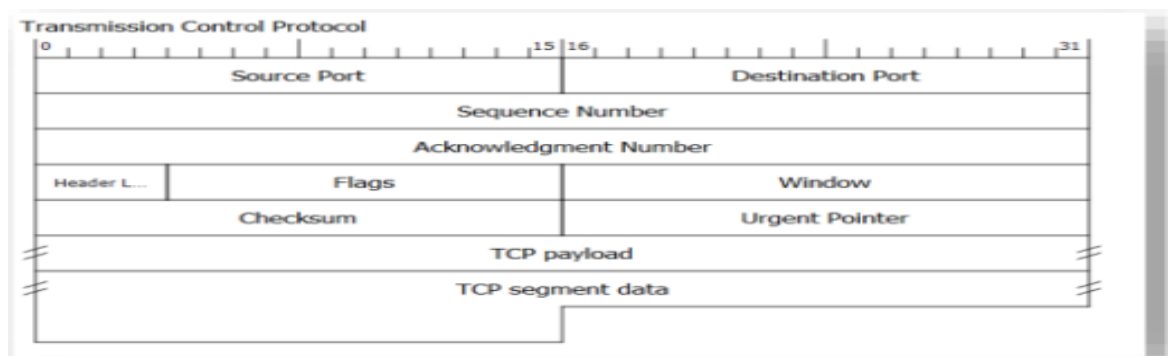
- Flags contain sub-fields such as: QR(Query/Response) which contains 0/1 depending on whether it is a query or a response.
- Opcode specifies the type of query the message is carrying.
- Truncation is set to 1 when message is truncated due to its length longer than the limit of transport medium used (512 bytes).
- Number of Questions is a 16-bit field which specifies the count of number of questions in Question Section.

- Number of answers RR is a 16-bit field which specifies the count of number of answer records in Answer Section of message.
- Queries contains the domain name and type of record (A, AAAA, MX, TXT, etc.) being resolved.



→TLSv1.3 (Transport Layer Security):

Transport Layer Security, or TLS, is a widely adopted security protocol designed to facilitate privacy and data security for communications over the Internet. Content Type specifies whether the content is Handshake, Application Data, Alert, Change Cipher Spec etc. Version field specifies the version of TLS being used. Here it is version 1.2. Length gives the length of packet including header. Payload contains the data of the packet and MAC is the Message authentication code.

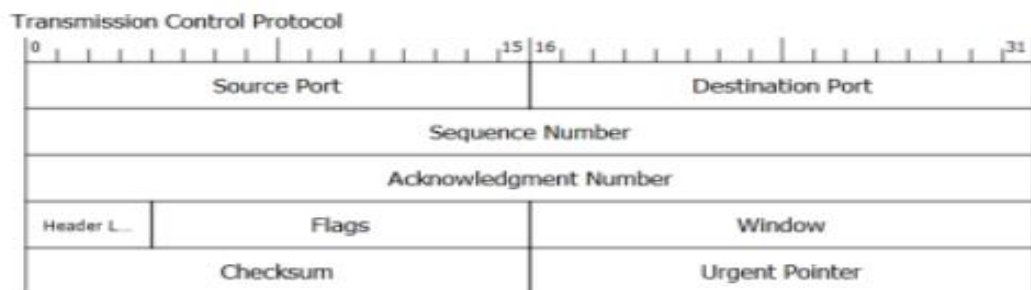


2) TRANSPORT LAYER:

→TRANSMISSION CONTROL PROTOCOL (TCP):

TCP is used for organizing data in a way that ensures the secure transmission between the server and client.

- Source Port Address: A 16-bit field that holds the port address of the application that is sending the data segment.
- Destination Port Address: A 16-bit field that holds the port address of the application in the host that is receiving the data segment.
- Sequence Number: A 32-bit field that holds the sequence number, the byte number of the first byte that is sent in that particular segment.
- Acknowledgement Number: A 32-bit field that holds the acknowledgement number, the byte number that the receiver expects to receive next.
- Control flags: These are 6 1-bit control bits that control connection establishment, connection termination, connection abortion, flow control, mode of transfer etc.
- Urgent pointer: This field (valid only if the URG control flag is set) is used to point to data that is urgently required that needs to reach the receiving process at the earliest.

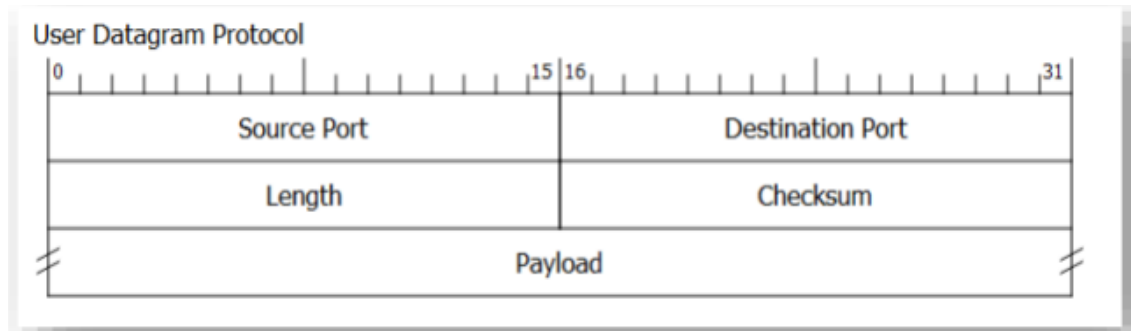


→USER DATAGRAM PROTOCOL (UDP):

User Datagram Protocol (UDP) is a communications protocol for time-sensitive applications like gaming, playing videos, or Domain Name System (DNS) lookups. Source Port: Source Port is a 2 Byte long field used to identify the port number of the source.

Destination Port: It is a 2 Byte long field, used to identify the port of the destined packet. Length: Length is the length of UDP including the header and the data. It is a 16-bits field. Checksum: Checksum is 2 Bytes long field. It is the 16-bit one's complement of the one's complement sum of the UDP header, the pseudo-

header of information from the IP header, and the data, padded with zero octets at the end (if necessary) to make a multiple of two octets.



3) INTERNET LAYER:

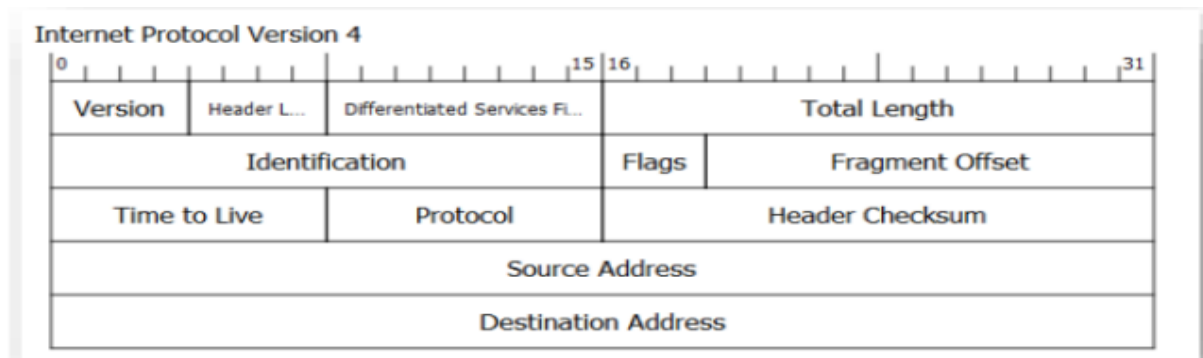
→INTERNET PROTOCOL VERSION 4:

Internet Protocol version 4 (IPv4) is the fourth version of the standard that routes Internet traffic and other packet-switched networks.

- VERSION: Version of the IP protocol (4 bits), which is 4 for IPv4
- Type of service: Low Delay, High Throughput, Reliability (8 bits)
- Total Length: Length of header + Data (16 bits), which has a minimum value 20 bytes and the maximum is 65,535 bytes.
- Identification: Unique Packet Id for identifying the group of fragments of a single IP datagram (16 bits).
- Flags: 3 flags of 1 bit each: reserved bit (must be zero), do not fragment flag, more fragments flag (same order).
- Fragment Offset: Represents the number of Data Bytes ahead of the particular fragment in the particular Datagram. Specified in terms of number of 8 bytes, which has the maximum value of 65,528 bytes.
- Time to live: Datagram's lifetime (8 bits), It prevents the datagram to loop through the network by restricting the

number of Hops taken by a Packet before delivering to the Destination.

- Protocol: Name of the protocol to which the data is to be passed (8 bits).
- Header Checksum: 16 bits header checksum for checking errors in the datagram header.
- Source IP address: 32 bits IP address of the sender.
- Destination IP address: 32 bits IP address of the receiver.

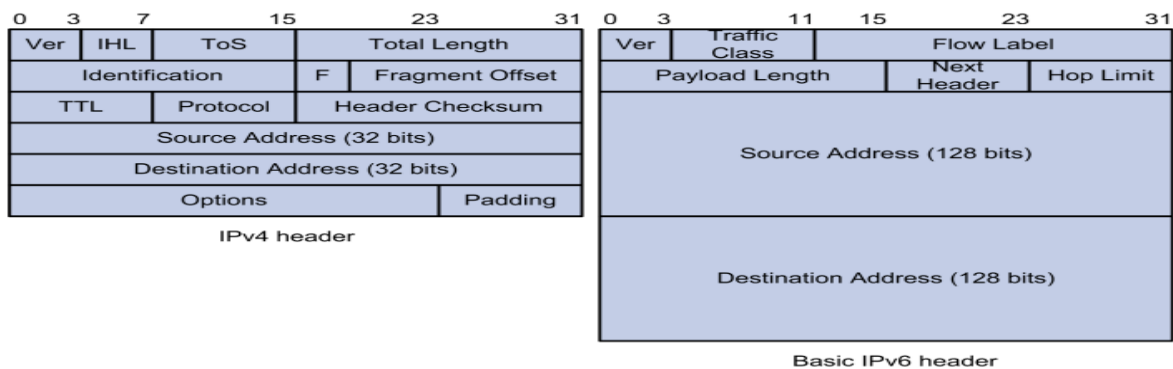


→INTERNET PROTOCOL VERSION 6(IPv6):

IPv6 is designed to replace IPv4, offering a larger address space and additional features like simplified packet header format and improved support for extensions and options. Packet Structure:

- Version: Indicates the IP version. For IPv6, this field is always set to 6.
- Traffic Class (8 bits): Similar to the Type of Service (ToS) in IPv4, this field is used for traffic prioritization. It can be used by routers and switches to determine the priority of the packet in case of congestion.
- Flow Label (20 bits): Used to identify packets that require special handling, such as those part of the same flow (e.g., video streaming, VoIP). It allows routers to recognize and handle all packets belonging to a particular flow without needing to examine the payload.
- Payload Length (16 bits): Specifies the length of the payload, i.e., the data following the header, in bytes. It can be up to 65,535 bytes. If it exceeds this, the payload length is set to zero, and the "Jumbo Payload" option is used in an extension header.

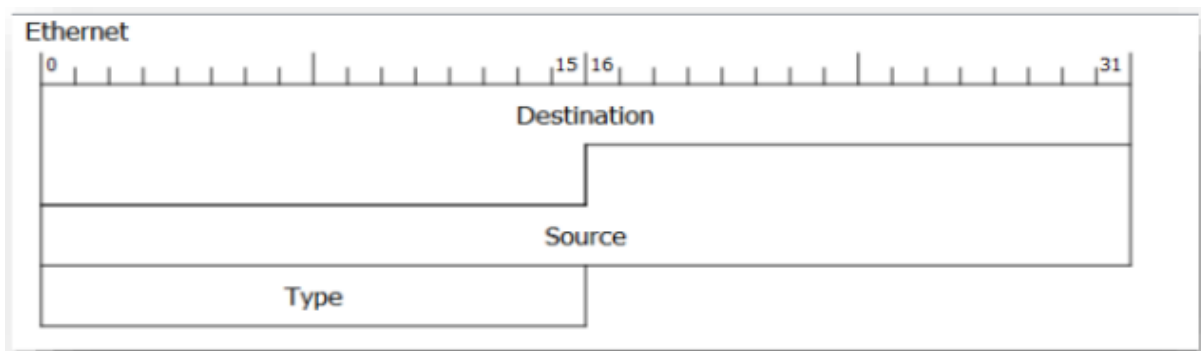
- Next Header (8 bits): Indicates the type of the header immediately following the IPv6 header. It could be a transport layer protocol (e.g., TCP, UDP) or extension header like routing or fragmentation headers. This field is similar to the Protocol field in IPv4.



4) LINK LAYER:

→ETHERNET

- Destination Address: This is a 6-Byte field that contains the MAC address (Media Access Control address) of the machine for which data is destined.
- Source address: This is a 6-Byte field that contains the MAC address (Media Access Control address) of the source machine.
- Type: Used to denote upper layer protocol that encapsulates the frame.



QUESTION 2:

DNS QUERY:

```
> Frame 189: 79 bytes on wire (632 bits), 79 bytes captured (632 bits) on interface \Device\NPF
> Ethernet II, Src: ChongqingFug_44:a1:fd (ac:d5:64:44:a1:fd), Dst: Dell_f0:ee:42 (c8:f7:50:f0:
> Internet Protocol Version 4, Src: 10.150.51.105, Dst: 172.17.1.1
> User Datagram Protocol, Src Port: 54380, Dst Port: 53
v Domain Name System (query)
  Transaction ID: 0xc9b7
  > Flags: 0x0100 Standard query
  Questions: 1
  Answer RRs: 0
  Authority RRs: 0
  Additional RRs: 0
  > Queries
    [Response In: 192]
```

The transaction ID of the DNS request is 0xc9b7. Response Flag tells us that this packet is a query and Opcode Flag specifies that the query type is standard. Questions in the pictures gives the number of queries which is 1. Answer RR is 0, Authority RR is 0 and Additional RR is also 0 (since it is a query but not a response).

DNS RESPONSE:

```
> Frame 191: 248 bytes on wire (1984 bits), 248 bytes captured (1984 bits) on interface \Device
> Ethernet II, Src: Dell_f0:ee:42 (c8:f7:50:f0:ee:42), Dst: ChongqingFug_44:a1:fd (ac:d5:64:44:
> Internet Protocol Version 4, Src: 172.17.1.1, Dst: 10.150.51.105
> User Datagram Protocol, Src Port: 53, Dst Port: 58083
v Domain Name System (response)
  Transaction ID: 0x84a1
  > Flags: 0x8180 Standard query response, No error
  Questions: 1
  Answer RRs: 4
  Authority RRs: 1
  Additional RRs: 0
  > Queries
  v Answers
    > teams.microsoft.com: type CNAME, class IN, cname teams.office.com
    > teams.office.com: type CNAME, class IN, cname tmc-g2.tm-4.office.com
    > tmc-g2.tm-4.office.com: type CNAME, class IN, cname teams-office-com.s-0005.dual-s-msedg
    > teams-office-com.s-0005.dual-s-msedge.net: type CNAME, class IN, cname s-0005.dual-s-ms
  v Authoritative nameservers
    > dual-s-msedge.net: type SOA, class IN, mname ns1.dual-s-msedge.net
    [Request In: 190]
    [Time: 0.001143000 seconds]
```

The transaction ID of the DNS request is 0x84a1. Questions in the pictures gives the number of queries which is 1. Answer RR is 4, Authority RR is 1 and Additional RR is also 0

TLSV1.3:

```
v Transport Layer Security
  v TLSv1.3 Record Layer: Application Data Protocol: Hypertext Transfer Protocol
    Opaque Type: Application Data (23)
    Version: TLS 1.2 (0x0303)
    Length: 6046
    Encrypted Application Data [...]: 3e026050e151eced7e1a9010e235e92680e5138776940f299aa364
    [Application Data Protocol: Hypertext Transfer Protocol]
```

We can observe that the TLS Version being used is TLS 1.3. The TLS handshake protocol is being used to authenticate the participants of the communication and negotiate an encryption algorithm. The length of the packet including header is 6046.

- ▼ Transport Layer Security
 - ▼ TLSv1 Record Layer: Handshake Protocol: Client Hello
 - Content Type: Handshake (22)
 - Version: TLS 1.0 (0x0301)
 - Length: 1820
 - > Handshake Protocol: Client Hello

TCP:

- ▼ Transmission Control Protocol, Src Port: 49510, Dst Port: 443, Seq: 0, Len: 0
 - Source Port: 49510
 - Destination Port: 443
 - [Stream index: 6]
 - [Stream Packet Number: 1]
 - > [Conversation completeness: Complete, WITH_DATA (31)]
 - [TCP Segment Len: 0]
 - Sequence Number: 0 (relative sequence number)
 - Sequence Number (raw): 1163590143
 - [Next Sequence Number: 1 (relative sequence number)]
 - Acknowledgment Number: 0
 - Acknowledgment number (raw): 0
 - 1000 = Header Length: 32 bytes (8)
 - > Flags: 0x002 (SYN)
 - Window: 65535
 - [Calculated window size: 65535]
 - Checksum: 0x7a0c [unverified]
 - [Checksum Status: Unverified]
 - Urgent Pointer: 0
 - > Options: (12 bytes), Maximum segment size, No-Operation (NOP), Window scale, No-Operation
 - > [Timestamps]

The Source port and Destination port in this case is 49510 and 443, respectively (of devices between which the communication is taking place). Sequence Number is 0 (The sequence number is a counter used to keep track of every byte sent outward by a host) and Acknowledgement Number is 0. Header length is 32 bytes. Window is 65535 and Urgent Pointer is 0 which means that no further bytes are urgent.

UDP (USER DATAGRAM PROTOCOL):

- User Datagram Protocol, Src Port: 53, Dst Port: 50940
 - Source Port: 53
 - Destination Port: 50940
 - Length: 197
 - Checksum: 0x518a [unverified]
 - [Checksum Status: Unverified]
 - [Stream index: 51]
 - [Stream Packet Number: 2]
 - > [Timestamps]
 - UDP payload (189 bytes)

The Source Port and the Destination Port are 53 and 50940, respectively. The Length of the packet is 197 and the Checksum Status is unverified. Stream index displays unique number for each stream, in this case it is 51.

IPv4:

- Internet Protocol Version 4, Src: 10.150.51.105, Dst: 52.123.128.14
 - 0100 = Version: 4
 - 0101 = Header Length: 20 bytes (5)
 - > Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
 - Total Length: 52
 - Identification: 0x0d49 (3401)
 - > 010. = Flags: 0x2, Don't fragment
 - ...0 0000 0000 0000 = Fragment Offset: 0
 - Time to Live: 128
 - Protocol: TCP (6)
 - Header Checksum: 0xfaf2 [validation disabled]
 - [Header checksum status: Unverified]
 - Source Address: 10.150.51.105
 - Destination Address: 52.123.128.14
 - [Stream index: 5]

The Header Length is 20 bytes and the Total Length of packet is 52. Fragment Offset is 0. Time to Live is 128 which means that packet can make at most 128 hops. The next level protocol to be used is TCP and source and destination addresses are 10.150.51.105 and 52.123.128.14 respectively.

ETHERNET:

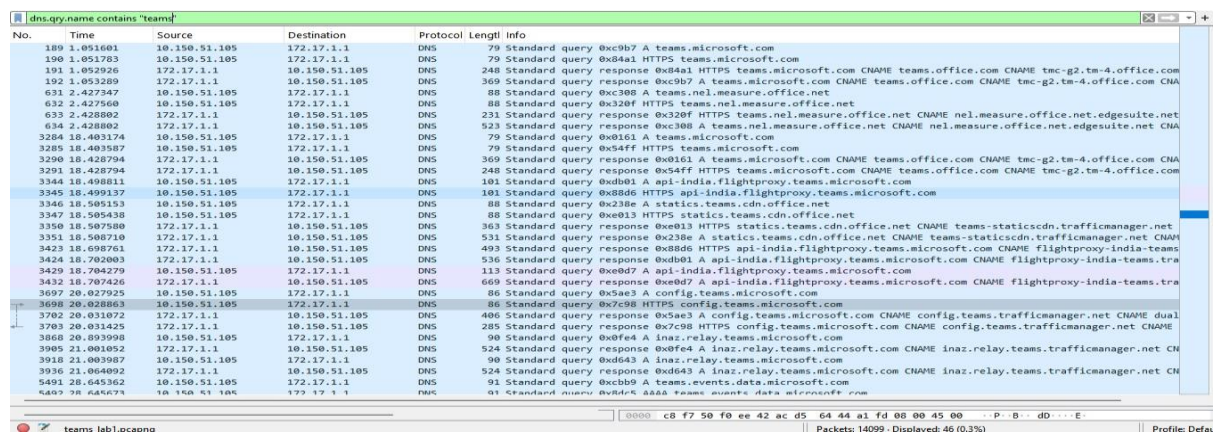
```
Ethernet II, Src: ChongqingFug_44:a1:fd (ac:d5:64:44:a1:fd), Dst: Dell_f0:ee:42 (c8:f7:50:f0:ee:42)
> Destination: Dell_f0:ee:42 (c8:f7:50:f0:ee:42)
> Source: ChongqingFug_44:a1:fd (ac:d5:64:44:a1:fd)
Type: IPv4 (0x0800)
[Stream index: 49]
```

THE SOURCE IS: ChongqingFug_44:a1:fd AND DESTINATION IS Dell_f0:ee:42.

QUESTION 3:

ESTABLISHING A CONNECTION:

DNS QUERY:



The screenshot shows a Wireshark packet capture of DNS traffic. The filter is 'dns.qry.name contains "teams"'. The packet list shows several DNS queries and responses. The packet details pane for packet 190 shows a standard query response for 'teams.microsoft.com' with CNAME and A records. The packet bytes pane shows the raw DNS data in hexadecimal and ASCII.

No.	Time	Source	Destination	Protocol	Length	Info
189	1.051601	10.150.51.105	172.17.1.1	DNS	79	Standard query 0xc9b7 A teams.microsoft.com
190	1.051783	10.150.51.105	172.17.1.1	DNS	79	Standard query response 0xc9b7 A teams.microsoft.com CNAME teams.office.com CNAME tmc-g2.tmc-4.office.com
191	1.052926	172.17.1.1	10.150.51.105	DNS	248	Standard query response 0xc9b7 A teams.microsoft.com CNAME teams.office.com CNAME tmc-g2.tmc-4.office.com
192	1.053209	172.17.1.1	10.150.51.105	DNS	369	Standard query response 0xc9b7 A teams.microsoft.com CNAME teams.office.com CNAME tmc-g2.tmc-4.office.com
631	2.427347	10.150.51.105	172.17.1.1	DNS	88	Standard query 0xc308 A teams.nel.measure.office.net
632	2.427568	10.150.51.105	172.17.1.1	DNS	88	Standard query 0xc308 A teams.nel.measure.office.net
633	2.428802	172.17.1.1	10.150.51.105	DNS	231	Standard query response 0xc308 A teams.nel.measure.office.net CNAME nel.measure.office.net.edgesuite.net
634	2.428802	172.17.1.1	10.150.51.105	DNS	523	Standard query response 0xc308 A teams.nel.measure.office.net CNAME nel.measure.office.net.edgesuite.net
3284	18.403174	10.150.51.105	172.17.1.1	DNS	79	Standard query 0xc0161 A teams.microsoft.com
3285	18.403587	10.150.51.105	172.17.1.1	DNS	79	Standard query 0xc0161 A teams.microsoft.com
3290	18.428794	172.17.1.1	10.150.51.105	DNS	369	Standard query response 0xc0161 A teams.microsoft.com CNAME teams.office.com CNAME tmc-g2.tmc-4.office.com
3291	18.428794	172.17.1.1	10.150.51.105	DNS	248	Standard query response 0xc0161 A teams.microsoft.com CNAME teams.office.com CNAME tmc-g2.tmc-4.office.com
3344	18.498811	10.150.51.105	172.17.1.1	DNS	101	Standard query 0xc0001 A api-india.flightproxy.teams.microsoft.com
3345	18.499137	10.150.51.105	172.17.1.1	DNS	101	Standard query 0xc0001 A api-india.flightproxy.teams.microsoft.com
3346	18.505153	10.150.51.105	172.17.1.1	DNS	88	Standard query 0xc238e A statics.teams.cdn.office.net
3347	18.505438	10.150.51.105	172.17.1.1	DNS	88	Standard query 0xc013 HTTPS statics.teams.cdn.office.net
3350	18.507589	172.17.1.1	10.150.51.105	DNS	363	Standard query response 0xc013 HTTPS statics.teams.cdn.office.net CNAME teams-staticscdn.trafficmanager.net
3351	18.508710	172.17.1.1	10.150.51.105	DNS	531	Standard query response 0xc238e A statics.teams.cdn.office.net CNAME teams-staticscdn.trafficmanager.net
3423	18.698761	172.17.1.1	10.150.51.105	DNS	493	Standard query response 0xc8d6 HTTPS api-india.flightproxy.teams.microsoft.com CNAME flightproxy-india.teams
3424	18.702003	172.17.1.1	10.150.51.105	DNS	536	Standard query response 0xc0001 A api-india.flightproxy.teams.microsoft.com CNAME flightproxy-india.teams
3429	18.704279	10.150.51.105	172.17.1.1	DNS	113	Standard query 0xc00d7 A api-india.flightproxy.teams.microsoft.com
3432	18.707426	172.17.1.1	10.150.51.105	DNS	669	Standard query response 0xc00d7 A api-india.flightproxy.teams.microsoft.com CNAME flightproxy-india.teams
3697	20.027925	10.150.51.105	172.17.1.1	DNS	86	Standard query 0xc5ae3 A config.teams.microsoft.com
3698	20.028863	10.150.51.105	172.17.1.1	DNS	86	Standard query 0xc7c98 HTTPS config.teams.microsoft.com
3702	20.031072	172.17.1.1	10.150.51.105	DNS	486	Standard query response 0xc5ae3 A config.teams.microsoft.com CNAME config.teams.trafficmanager.net CNAME dual
3703	20.031425	172.17.1.1	10.150.51.105	DNS	285	Standard query response 0xc7c98 HTTPS config.teams.microsoft.com CNAME config.teams.trafficmanager.net CNAME
3868	20.093990	10.150.51.105	172.17.1.1	DNS	90	Standard query 0xc0fe4 A inaz.relay.teams.microsoft.com
3905	21.001052	172.17.1.1	10.150.51.105	DNS	524	Standard query response 0xc0fe4 A inaz.relay.teams.microsoft.com CNAME inaz.relay.teams.trafficmanager.net
3918	21.003987	10.150.51.105	172.17.1.1	DNS	90	Standard query 0xc0d43 A inaz.relay.teams.microsoft.com
3936	21.004002	172.17.1.1	10.150.51.105	DNS	524	Standard query response 0xc0d43 A inaz.relay.teams.microsoft.com CNAME inaz.relay.teams.trafficmanager.net
5491	28.645362	10.150.51.105	172.17.1.1	DNS	91	Standard query 0xcbb9 A teams.events.data.microsoft.com
6409	28.645673	10.150.51.105	172.17.1.1	DNS	91	Standard query 0xcbb9 A teams.events.data.microsoft.com

The client (192.168.0.139) first sends DNS queries for Teams-related domains such as teams.microsoft.com and teams.events.data.microsoft.com. The DNS server responds with CNAME and A records mapping these hostnames to Microsoft IP addresses (e.g., 52.123.129.14, 52.123.128.14).

TLS HANDSHAKE:

TO SECURE A CONNECTION THE CLIENT INITIATES A HANDSHAKE:

tls.handshake and (ip.addr==52.123.129.14 or ip.addr==52.123.128.14)						
No.	Time	Source	Destination	Protocol	Length	Info
206	1.055917	10.150.51.105	52.123.128.14	TLSv1.3	483	Client Hello (SNI=teams.microsoft.com)
208	1.056322	10.150.51.105	52.123.128.14	TLSv1.3	419	Client Hello (SNI=teams.microsoft.com)
210	1.056729	10.150.51.105	52.123.128.14	TLSv1.3	451	Client Hello (SNI=teams.microsoft.com)
212	1.057118	10.150.51.105	52.123.128.14	TLSv1.3	451	Client Hello (SNI=teams.microsoft.com)
221	1.112830	52.123.128.14	10.150.51.105	TLSv1.3	153	Hello Retry Request, Change Cipher Spec
222	1.112830	52.123.128.14	10.150.51.105	TLSv1.3	153	Hello Retry Request, Change Cipher Spec
223	1.112978	52.123.128.14	10.150.51.105	TLSv1.3	153	Hello Retry Request, Change Cipher Spec
224	1.115605	10.150.51.105	52.123.128.14	TLSv1.3	693	Change Cipher Spec, Client Hello (SNI=teams.microsoft.com)
226	1.118479	10.150.51.105	52.123.128.14	TLSv1.3	661	Change Cipher Spec, Client Hello (SNI=teams.microsoft.com)
227	1.121251	10.150.51.105	52.123.128.14	TLSv1.3	693	Change Cipher Spec, Client Hello (SNI=teams.microsoft.com)
230	1.127028	52.123.128.14	10.150.51.105	TLSv1.3	153	Hello Retry Request, Change Cipher Spec
231	1.130028	10.150.51.105	52.123.128.14	TLSv1.3	725	Change Cipher Spec, Client Hello (SNI=teams.microsoft.com)
233	1.158282	52.123.128.14	10.150.51.105	TLSv1.3	2846	Server Hello
236	1.158463	52.123.128.14	10.150.51.105	TLSv1.3	2846	Server Hello
242	1.158463	52.123.128.14	10.150.51.105	TLSv1.3	1450	Server Hello
277	1.180729	52.123.128.14	10.150.51.105	TLSv1.3	1450	Server Hello
3319	18.435594	10.150.51.105	52.123.129.14	TLSv1.2	483	Client Hello (SNI=teams.microsoft.com)
5096	26.858936	10.150.51.105	52.123.129.14	TLSv1.2	483	Client Hello (SNI=teams.microsoft.com)
6135	31.056082	10.150.51.105	52.123.129.14	TLSv1.2	483	Client Hello (SNI=teams.microsoft.com)
8086	41.058952	10.150.51.105	52.123.129.14	TLSv1.2	483	Client Hello (SNI=teams.microsoft.com)
9016	49.368546	10.150.51.105	52.123.129.14	TLSv1.2	419	Client Hello (SNI=teams.microsoft.com)
12288	61.057565	10.150.51.105	52.123.129.14	TLSv1.2	451	Client Hello (SNI=teams.microsoft.com)
12794	63.095423	10.150.51.105	52.123.129.14	TLSv1.2	451	Client Hello (SNI=teams.microsoft.com)
13531	66.383686	10.150.51.105	52.123.129.14	TLSv1.2	451	Client Hello (SNI=teams.microsoft.com)

After resolving the hostname, the client initiates a TCP connection to the Teams server on port 443. SYN (Client → Server), SYN/ACK (Server → Client), ACK (Client → Server). This establishes the TCP connection successfully.

The client then performs a TLS 1.2 handshake with the Teams server. Client Hello, including the Server Name Indication (SNI) = teams.microsoft.com. The server replies with Server Hello, Certificate, and key exchange messages. This secures the connection with encryption.

Encrypted Application Data:

253	1.166672	10.150.51.105	52.123.128.14	TCP	1450	49511 → 443 [ACK] Seq=2599 Ack=6149 Win=65280 Len=1396 [TCP PDU reassembled in 255]
254	1.166672	10.150.51.105	52.123.128.14	TCP	1450	49511 → 443 [ACK] Seq=3995 Ack=6149 Win=65280 Len=1396 [TCP PDU reassembled in 255]
255	1.166672	10.150.51.105	52.123.128.14	TLSv1.3	340	Application Data
256	1.166897	10.150.51.105	52.123.128.14	TCP	1450	49511 → 443 [ACK] Seq=5677 Ack=6149 Win=65280 Len=1396 [TCP PDU reassembled in 258]
257	1.166897	10.150.51.105	52.123.128.14	TCP	1450	49511 → 443 [ACK] Seq=7073 Ack=6149 Win=65280 Len=1396 [TCP PDU reassembled in 258]
258	1.166897	10.150.51.105	52.123.128.14	TLSv1.3	482	Application Data
259	1.167053	10.150.51.105	52.123.128.14	TCP	1450	49511 → 443 [ACK] Seq=8897 Ack=6149 Win=65280 Len=1396 [TCP PDU reassembled in 261]
260	1.167053	10.150.51.105	52.123.128.14	TCP	1450	49511 → 443 [ACK] Seq=10293 Ack=6149 Win=65280 Len=1396 [TCP PDU reassembled in 261]
261	1.167053	10.150.51.105	52.123.128.14	TLSv1.3	648	Application Data
262	1.167188	10.150.51.105	52.123.128.14	TCP	1450	49511 → 443 [ACK] Seq=12283 Ack=6149 Win=65280 Len=1396 [TCP PDU reassembled in 264]
263	1.167188	10.150.51.105	52.123.128.14	TCP	1450	49511 → 443 [ACK] Seq=13679 Ack=6149 Win=65280 Len=1396 [TCP PDU reassembled in 264]
264	1.167188	10.150.51.105	52.123.128.14	TLSv1.3	612	Application Data
265	1.167343	10.150.51.105	52.123.128.14	TCP	1450	49511 → 443 [ACK] Seq=15633 Ack=6149 Win=65280 Len=1396 [TCP PDU reassembled in 267]
266	1.167343	10.150.51.105	52.123.128.14	TCP	1450	49511 → 443 [ACK] Seq=17029 Ack=6149 Win=65280 Len=1396 [TCP PDU reassembled in 267]
267	1.167343	10.150.51.105	52.123.128.14	TLSv1.3	394	Application Data
268	1.174326	52.123.128.14	10.150.51.105	TCP	1450	443 → 49513 [ACK] Seq=1496 Ack=2433 Win=26752 Len=1396 [TCP PDU reassembled in 271]
269	1.174326	52.123.128.14	10.150.51.105	TCP	182	443 → 49513 [PSH, ACK] Seq=2892 Ack=2433 Win=26752 Len=128 [TCP PDU reassembled in 271]
270	1.174326	52.123.128.14	10.150.51.105	TCP	2846	443 → 49513 [ACK] Seq=3020 Ack=2433 Win=26752 Len=2792 [TCP PDU reassembled in 271]
271	1.174326	52.123.128.14	10.150.51.105	TLSv1.3	391	Application Data

Following the TLS handshake, Teams begins exchanging TLS Application Data packets. These carry the actual Teams traffic such as signaling, chat, and possibly audio/video. Because no UDP/STUN/RTP traffic is observed in this capture, Teams is operating in TCP fallback mode over port 443.

The observed sequence is:

DNS Resolution → TCP 3-Way Handshake → TLS Handshake (SNI = teams.microsoft.com) → Encrypted TLS Application Data. This confirms that Teams traffic in this environment is fully encrypted and transported over TCP port 443.

QUESTION 4:

When using TEAMS and performing several activities on it, several protocols work together to ensure the process is secure, efficient, and reliable. Here's how these protocols contribute to the overall functionality of MICROSOFT TEAMS:

TLS (Transport Layer Security) TLS is the underlying security protocol that powers HTTPS. It provides encryption, authentication, and data integrity for the connection between the client and server.

- Encrypts the data sent between the client and server, preventing unauthorized access to the messages and data.
- Establishes a secure session with the server, negotiating encryption keys that will be used to protect sending process.
- Allows for faster reconnections by reusing session keys, improving the user experience during long or interrupted calls.

DNS (Domain Name System) DNS is used to resolve Team's domain names into IP addresses that the client can use to connect to the server.

- Converts the human-readable domain names into IP addresses, allowing the client to establish a connection with Microsoft servers.
- DNS can be used to distribute the callrequest across multiple servers, optimizing load balancing and improving callreliability.

TCP (Transmission Control Protocol) TCP is the transport protocol used by HTTPS and HTTP/2. It ensures that the video data is reliably delivered to the server, even if network conditions are suboptimal.

- Guarantees that all data packets are delivered in the correct order and without errors, which is crucial for maintaining the integrity of the video call.
- Manages the flow of data to prevent network congestion, which helps maintain consistent chats.
- Automatically retransmits lost packets, ensuring that the entire message is received by the server without corruption.

UDP (User Datagram Protocol) It operates over UDP instead of TCP. This allows for faster, more efficient communication, particularly in scenarios where low latency is critical.

- UDP reduces the overhead associated with connection management, leading to faster messages and calls.

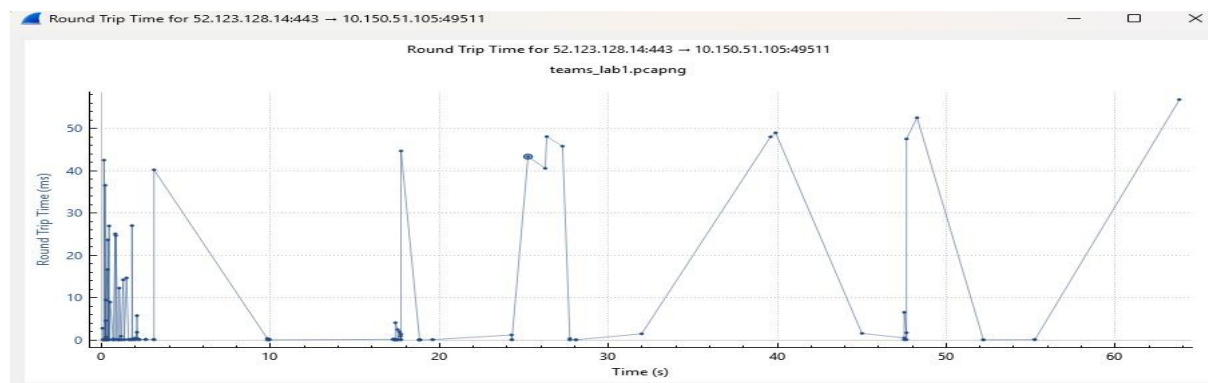
- Handles packet loss and reordering in a way that is more forgiving and efficient compared to TCP, making it suitable for video calls.

No.	Time	Source	Destination	Protocol	Length	Info
189	1.051691	10.150.51.105	172.17.1.1	DNS	79	Standard query 0xc9b7 A teams.microsoft.com
190	1.051783	10.150.51.105	172.17.1.1	DNS	79	Standard query 0xc9b7 A teams.microsoft.com
191	1.052926	172.17.1.1	10.150.51.105	DNS	248	Standard query response 0xc9b7 A teams.microsoft.com CNAME teams.office.com CNAME tmc-g2.tm-4.office.com
192	1.053289	172.17.1.1	10.150.51.105	DNS	369	Standard query response 0xc9b7 A teams.microsoft.com CNAME teams.office.com CNAME tmc-g2.tm-4.office.com
631	2.422347	10.150.51.105	172.17.1.1	DNS	88	Standard query 0xc308 A teams.nel.measure.office.net
632	2.422560	10.150.51.105	172.17.1.1	DNS	88	Standard query 0xc30f HTTPS teams.nel.measure.office.net
633	2.428802	172.17.1.1	10.150.51.105	DNS	231	Standard query response 0xc30f HTTPS teams.nel.measure.office.net CNAME nel.measure.office.net.edgesuite.net
634	2.428802	172.17.1.1	10.150.51.105	DNS	523	Standard query response 0xc308 A teams.nel.measure.office.net CNAME nel.measure.office.net.edgesuite.net
3284	18.403174	10.150.51.105	172.17.1.1	DNS	79	Standard query 0xc9161 A teams.microsoft.com
3285	18.403587	10.150.51.105	172.17.1.1	DNS	79	Standard query 0xc9161 A teams.microsoft.com
3290	18.428794	172.17.1.1	10.150.51.105	DNS	369	Standard query response 0xc9161 A teams.microsoft.com CNAME teams.office.com CNAME tmc-g2.tm-4.office.com
3291	18.428794	172.17.1.1	10.150.51.105	DNS	248	Standard query response 0xc9161 A teams.microsoft.com CNAME teams.office.com CNAME tmc-g2.tm-4.office.com
3344	18.498811	10.150.51.105	172.17.1.1	DNS	101	Standard query 0xcdb01 A api-india.flightproxy.teams.microsoft.com
3345	18.499137	10.150.51.105	172.17.1.1	DNS	101	Standard query 0xcdb01 A api-india.flightproxy.teams.microsoft.com
3346	18.505153	10.150.51.105	172.17.1.1	DNS	88	Standard query 0xcdb01 A api-india.flightproxy.teams.microsoft.com
3347	18.505438	10.150.51.105	172.17.1.1	DNS	88	Standard query 0xcdb01 A api-india.flightproxy.teams.microsoft.com
3350	18.507500	172.17.1.1	10.150.51.105	DNS	363	Standard query response 0xcdb01 A api-india.flightproxy.teams.microsoft.com
3351	18.508710	172.17.1.1	10.150.51.105	DNS	531	Standard query response 0xcdb01 A api-india.flightproxy.teams.microsoft.com
3423	18.698761	172.17.1.1	10.150.51.105	DNS	493	Standard query response 0xcdb01 A api-india.flightproxy.teams.microsoft.com
3424	18.702003	172.17.1.1	10.150.51.105	DNS	536	Standard query response 0xcdb01 A api-india.flightproxy.teams.microsoft.com
3697	20.027925	10.150.51.105	172.17.1.1	DNS	86	Standard query 0xc9a3 A config.teams.microsoft.com

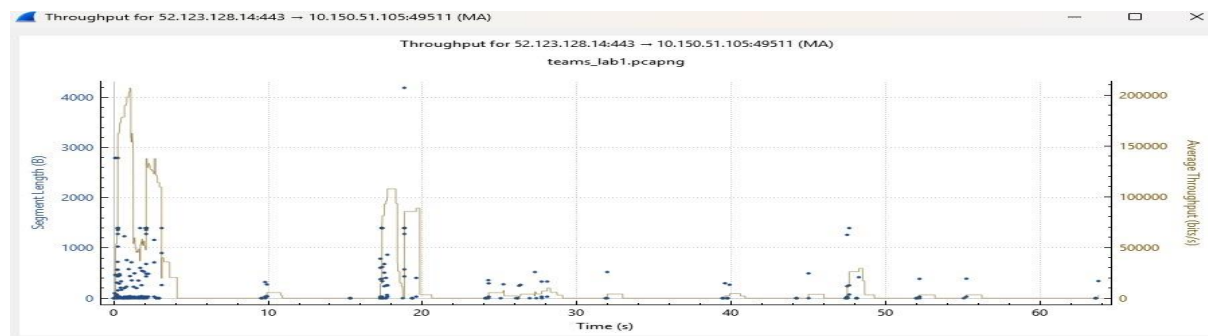
QUESTION 5:

	THROUGHPUT(bytes/s)	RTT max (ms)	PACKET SIZE (B)	PACKETS LOST	NUMBER OF UDP AND TCP PACKETS	NUMBER OF RESPONSES RECEIVED W.R.T REQUEST SENT
AFTERNOON	120k	48	811.45	0(lost), 6(retransmit)	2279(TCP) 94(UDP)	1
EVENING	48k	53	197.57	0(lost), 1(retransmit)	808(TCP), 46(UDP)	1

RTT:



THROUGHPUT:



PACKET SIZE:

Wireshark · Packet Lengths · teams_lab1.pcapng

Topic / Item	Count	Average	Min Val	Max Val	Rate (ms)	Percent	Burst Rate	Burst Start
Packet Lengths	14099	197.57	42	7034	0.2019	100%	1.2200	1.158
0-19	0	-	-	-	0.0000	0.00%	-	-
20-39	0	-	-	-	0.0000	0.00%	-	-
40-79	6938	57.54	42	79	0.0993	49.21%	0.5700	1.158
80-159	3861	105.27	80	159	0.0553	27.38%	0.3200	29.283
160-319	1453	204.28	160	319	0.0208	10.31%	0.1800	16.180
320-639	848	461.41	320	637	0.0121	6.01%	0.2100	19.457
640-1279	364	867.46	640	1279	0.0052	2.58%	0.2400	8.116
1280-2559	603	1439.76	1287	2466	0.0086	4.28%	0.3400	20.995
2560-5119	30	3171.73	2846	4242	0.0004	0.21%	0.0600	27.520
5120 and greater	2	6336.00	5638	7034	0.0000	0.01%	0.0100	22.217

netstat -s

```

Administrator: Windows PowerShell
PS C:\Windows\system32> netstat -s

IPv4 Statistics
Packets Received          = 2005546
Received Header Errors    = 31
Received Address Errors   = 1629
Datagrams Forwarded       = 0
Unknown Protocols Received = 0
Received Packets Discarded = 203721
Received Packets Delivered = 1987763
Output Requests           = 277207
Routing Discards          = 0
Discarded Output Packets  = 3378
Output Packet No Route    = 117
Reassembly Required       = 85
Reassembly Successful     = 11
Reassembly Failures       = 0
Datagrams Successfully Fragmented = 0
Datagrams Failing Fragmentation = 0
Fragments Created        = 0

IPv6 Statistics
Packets Received          = 908190
Received Header Errors    = 0
Received Address Errors   = 474841
Datagrams Forwarded       = 0
Unknown Protocols Received = 0
Received Packets Discarded = 20352
Received Packets Delivered = 233792
Output Requests           = 57911
Routing Discards          = 0
Discarded Output Packets  = 3593
Output Packet No Route    = 0
Reassembly Required       = 4121
Reassembly Successful     = 403
Reassembly Failures       = 0
Datagrams Successfully Fragmented = 0
Datagrams Failing Fragmentation = 0
Fragments Created        = 0
  
```

QUESTION 6:

Wireshark · Endpoints · teams_lab1.pcapng

Endpoint Settings

☐ Name resolution

☒ Limit to display filter

Copy

Map

Protocol

☐ Bluetooth

☐ BPv7

☐ DCCP

☒ Ethernet

☐ FC

☐ FDDI

☐ IEEE 802.11

☐ IEEE 802.15.4

☒ IPv4

☒ IPv6

Filter list for specific type

Ethernet - 2

IPv4 - 9

IPv6

TCP - 33

UDP - 25

Address	Packets	Bytes	Total Packets	Percent Filtered	Tx Packets	Tx Bytes	Rx Packets	Rx Bytes	Country	City	Latitude	Longitude	AS Number	AS Organization
10.150.51.105	854	412 kB	4,515	18.91%	416	272 kB	438	141 kB						
13.69.116.109	1	273 bytes	28	3.57%	0	0 bytes	1	273 bytes						
23.36.177.136	44	11 kB	44	100.00%	24	6 kB	20	4 kB						
23.36.177.139	58	12 kB	58	100.00%	32	4 kB	26	9 kB						
52.113.158.0	1	371 bytes	94	1.06%	0	0 bytes	1	371 bytes						
52.123.128.14	631	358 kB	631	100.00%	321	119 kB	310	239 kB						
52.123.129.14	70	18 kB	70	100.00%	36	2 kB	34	16 kB						
52.123.164.172	1	505 bytes	349	0.29%	0	0 bytes	1	505 bytes						
172.17.1.1	48	12 kB	103	46.60%	25	10 kB	23	2 kB						

MS Teams is a real-time collaboration app (video calls, chat, file sharing). It never relies on a single server. Instead, it uses Microsoft's global cloud infrastructure (Azure + Office365 + CDN). So in Wireshark traces we see:

Signaling traffic → to Microsoft login & Teams servers (usually HTTPS, TCP/443). Media traffic (audio/video) → to Microsoft's media relays (TURN/ICE servers over UDP). CDN traffic → for file uploads/downloads and images (OneDrive/SharePoint, via HTTPS).

Multiple IP addresses (belonging to Microsoft, Akamai, or Azure datacenters). Content does not come from one location.

WHY MULTIPLE SOURCES:

Content Delivery Networks (CDNs): Applications use CDNs to deliver content from geographically distributed servers, reducing latency and improving performance.

Load Balancing: Traffic is distributed across multiple servers to enhance reliability and scalability, leading to multiple IP addresses in network traces.

Redundancy and Failover: Applications route traffic through different servers to ensure continuous availability, especially during server failures or overloads.

Microservices Architecture: Applications often break down functionality into independent services, each potentially hosted on different servers, resulting in multiple IPs.

Conclusion: For MS Teams, content comes from multiple sources, because Teams uses distributed Azure datacenters, media relays, and CDNs to ensure reliability and low latency.

QUESTION 7:

In MS Teams, traffic is carried over HTTP (TLS-encrypted HTTP). Using Wireshark, we captured flows to teams.microsoft.com and graph.microsoft.com. These flows encapsulate HTTP GET and POST transactions, but the actual request/response headers and content are encrypted, so we cannot directly view fields such as request URL, User-Agent, or content type. Based on Teams' functionality: GET requests are likely used when fetching chat history, meeting details, and presence information. POST requests are used when sending chat messages, uploading files, or joining meetings. The destination port is 443, and the client/server IPs correspond to Microsoft Azure datacenters.

Comparison: GET is used for data retrieval (safe, idempotent). POST is used for sending/modifying data (not idempotent, carries request body). Therefore, although the exact HTTP message fields cannot be shown due to encryption, the observed TLS flows confirm that Teams relies heavily on HTTP GET/POST transactions behind the scenes.

tcp.port == 443 && (ip.dst == 52.123.129.14 or ip.dst == 52.123.128.14)						
Packet list Display filter Enter a display filter ... Filter						
Options: Narrow & Wide Case sensitive Backwards Multiple occurrences						
No.	Time	Source	Destination	Protocol	Length	Info
193	1.053745	10.150.51.105	52.123.128.14	TCP	66	49510 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
194	1.053901	10.150.51.105	52.123.128.14	TCP	66	49511 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
195	1.054040	10.150.51.105	52.123.128.14	TCP	66	49512 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
196	1.054151	10.150.51.105	52.123.128.14	TCP	66	49513 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
199	1.055411	10.150.51.105	52.123.128.14	TCP	54	49510 → 443 [ACK] Seq=1 Ack=1 Win=65280 Len=0
200	1.055435	10.150.51.105	52.123.128.14	TCP	54	49512 → 443 [ACK] Seq=1 Ack=1 Win=65280 Len=0
203	1.055877	10.150.51.105	52.123.128.14	TCP	54	49511 → 443 [ACK] Seq=1 Ack=1 Win=65280 Len=0
204	1.055901	10.150.51.105	52.123.128.14	TCP	54	49513 → 443 [ACK] Seq=1 Ack=1 Win=65280 Len=0
205	1.055917	10.150.51.105	52.123.128.14	TCP	1450	49512 → 443 [ACK] Seq=1 Ack=1 Win=65280 Len=1396 [TCP PDU reassembled in 206]
206	1.055917	10.150.51.105	52.123.128.14	TLSv1.3	483	Client Hello (SNI=teams.microsoft.com)
207	1.056322	10.150.51.105	52.123.128.14	TCP	1450	49510 → 443 [ACK] Seq=1 Ack=1 Win=65280 Len=1396 [TCP PDU reassembled in 208]
208	1.056322	10.150.51.105	52.123.128.14	TLSv1.3	419	Client Hello (SNI=teams.microsoft.com)
209	1.056729	10.150.51.105	52.123.128.14	TCP	1450	49513 → 443 [ACK] Seq=1 Ack=1 Win=65280 Len=1396 [TCP PDU reassembled in 210]
210	1.056729	10.150.51.105	52.123.128.14	TLSv1.3	451	Client Hello (SNI=teams.microsoft.com)
211	1.057118	10.150.51.105	52.123.128.14	TCP	1450	49511 → 443 [ACK] Seq=1 Ack=1 Win=65280 Len=1396 [TCP PDU reassembled in 212]
212	1.057118	10.150.51.105	52.123.128.14	TLSv1.3	451	Client Hello (SNI=teams.microsoft.com)
224	1.115605	10.150.51.105	52.123.128.14	TLSv1.3	693	Change Cipher Spec, Client Hello (SNI=teams.microsoft.com)
226	1.116479	10.150.51.105	52.123.128.14	TLSv1.3	661	Change Cipher Spec, Client Hello (SNI=teams.microsoft.com)
227	1.121251	10.150.51.105	52.123.128.14	TLSv1.3	693	Change Cipher Spec, Client Hello (SNI=teams.microsoft.com)
231	1.130028	10.150.51.105	52.123.128.14	TLSv1.3	725	Change Cipher Spec, Client Hello (SNI=teams.microsoft.com)
234	1.158345	10.150.51.105	52.123.128.14	TCP	54	49511 → 443 [ACK] Seq=2433 Ack=2892 Win=65280 Len=0

QUESTION 8:

dns and frame contains "iitg"						
No.	Time	Source	Destination	Protocol	Length	Info
2194	8.519928	10.150.51.105	172.17.1.1	DNS	70	Standard query 0x2954 A iitg.ac.in
2195	8.520298	10.150.51.105	172.17.1.1	DNS	70	Standard query 0xdd87 HTTPS iitg.ac.in
2197	8.521686	172.17.1.1	10.150.51.105	DNS	120	Standard query response 0xdd87 HTTPS iitg.ac.in SOA dns2.iitg.ac.in
2198	8.521686	172.17.1.1	10.150.51.105	DNS	156	Standard query response 0x2954 A iitg.ac.in A 172.17.0.22 NS dns2.iitg.ac.in NS dns1.iitg.ac.in A 172.17.1.1 A 172.17.1.1
2449	8.762832	10.150.51.105	172.17.1.1	DNS	75	Standard query 0x8b74 A wpad.iitg.ac.in
2450	8.765136	172.17.1.1	10.150.51.105	DNS	125	Standard query response 0x8b74 No such name A wpad.iitg.ac.in SOA dns2.iitg.ac.in
4309	16.779364	10.150.51.105	172.17.1.1	DNS	75	Standard query 0x81a6 A wpad.iitg.ac.in
4311	16.780890	172.17.1.1	10.150.51.105	DNS	125	Standard query response 0x81a6 No such name A wpad.iitg.ac.in SOA dns2.iitg.ac.in

IP ADDRESS OF IITG: 172.17.0.22

IP ADDRESS OF DNS: 172.17.1.1

NAME SERVER FOR DOMAIN: dns1.iitg.ac.in, dns2.iitg.ac.in

To find IP address to nslookup <name_server>

Transport protocol used is TCP

Port numbers: source:62070 destination:443

Each dns query gets a single dns response.

When accessing the domain www.iitg.ac.in, the initial packets captured in Wireshark show DNS queries generated by the client to resolve the hostname into an IP address. These queries are sent using the DNS protocol over UDP (port 53) to the configured DNS resolver. The corresponding DNS responses provide the IP address of the IIT Guwahati web server, along with details of the authoritative name servers for the domain iitg.ac.in.

After the DNS resolution is completed, the subsequent communication shifts to TLS-encrypted traffic (over TCP port 443), since the IIT Guwahati website enforces HTTPS connections. At this stage, the client directly communicates with the resolved IP address of the web server without issuing additional DNS lookups for the same domain. This behavior is due to DNS caching at the client or local resolver, which stores the earlier responses for reuse within their Time-To-Live (TTL) validity period.

It is also observed that the resolved server IPs may differ slightly across sessions depending on load balancing or the DNS resolver being used. However, within a single

session, once the initial DNS resolution is done, the communication proceeds almost entirely over TLS packets exchanged with the previously resolved IP address.

In summary:

DNS activity is visible only during the initial hostname resolution. Subsequent communication uses TLS with the resolved server IP, without repeated DNS lookups. Caching ensures efficiency, and DNS lookups will only reoccur if the TTL expires or if a different service endpoint under iitg.ac.in is requested.

CACHING:

No.	Time	Source	Destination	Protocol	Length	Info
688	3.744549	10.150.51.105	172.17.0.22	TLSv1.3	777	Client Hello (SNI=iitg.ac.in)
1084	4.853680	10.150.51.105	172.17.0.22	TLSv1.3	713	Client Hello (SNI=iitg.ac.in)

nslookup -type=NS iitg.ac.in

```
C:\WINDOWS\system32>nslookup iitg.ac.in
Server:      UnKnown
Address:     172.17.1.1

Name:       iitg.ac.in
Address:    172.17.0.22

C:\WINDOWS\system32>nslookup -type=NS iitg.ac.in
Server:      UnKnown
Address:     172.17.1.1

iitg.ac.in      nameserver = dns2.iitg.ac.in
iitg.ac.in      nameserver = dns1.iitg.ac.in
dns1.iitg.ac.in internet address = 172.17.1.1
dns2.iitg.ac.in internet address = 172.17.1.2

C:\WINDOWS\system32>
```